

Weiyan Shi (she/her/hers)

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POSITIONS	Assistant Professor , Northeastern University ECE & Khoury Computer Science	2024 – Present
	Postdoc , Stanford University Advisor: Diyi Yang	2023 – 2024
	Data Scientist , [24]7.ai	2016 – 2018
EDUCATION	Columbia University Ph.D. in Computer Science Advisor: Zhou Yu	2021 – 2023
	University of California, Davis Ph.D. in Computer Science (transferred with advisor)	2018 – 2021
	University of California, Berkeley M.A. in Statistics	2015 – 2016
	Renmin University of China B.S. in Mathematics and Applied Mathematics	2011 – 2015
SELECTED	Best Paper, ICML DL4C Workshop	2026
AWARDS	Outstanding Paper, NeurIPS SEA Workshop	2025
	Schmidt Sciences AI2050 Early Career Fellowship	2025
	MIT Technology Review 35 Innovators under 35	2024
	Best Social Impact Paper, ACL	2024
	Outstanding Paper, ACL	2024
	Best Paper Nomination, ACL	2019
	Department Citation (top 1), UC Berkeley Statistics	2016
	Speaker at the Department Commencement (top 1), UC Berkeley Statistics	2016

JOURNAL PUBLICATIONS

3. [SocialFusion: Addressing Social Degradation in Pre-trained Vision-Language Models](#)
Hamza Tahboub, [Weiyan Shi](#), Gang Hua, Huaizu Jiang
Transactions on Machine Learning Research (TMLR), 2026
2. [Labeling Messages as AI-Generated Does Not Reduce Their Persuasive Effects](#)
Isabel Gallegos, Chen Shani, [Weiyan Shi](#), Federico Bianchi, Izzy Gainsburg, Dan Jurafsky, Robb Willer
PNAS Nexus, 2026

1. **Human-Level Play in the Game of Diplomacy by Combining Language Models with Strategic Reasoning**
FAIR, Anton Bakhtin*, Noam Brown*, Emily Dinan*, Gabriele Farina, Colin Flaherty*, Daniel Fried, Andrew Goff, Jonathan Gray*, Hengyuan Hu*, Athul Paul Jacob*, Mojtaba Komeili, Karthik Konath, Minae Kwon, Adam Lerer*, Mike Lewis*, Alexander H. Miller*, Sasha Mitts, Adithya Renduchintala*, Stephen Roller, Dirk Rowe, Weiyan Shi*, Joe Spisak, Alexander Wei, David Wu*, Hugh Zhang*, Markus Zijlstra (* core member; authors listed alphabetically)
Science, 2022
[Meta AI blog] [Science News] **Coverage:** [New York Times front page] [Washington Post] [Economist] [MIT Tech Review] [Forbes]

CONFERENCE PUBLICATIONS

43. **Which Metrics Save the Most Human Annotation? Prediction-Powered Evaluation and Meta-Evaluation**
Mingqi Gao, Anthony Sicilia, Weiyan Shi
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026
42. **One Frozen Simulator Is Not Enough: Simulator Collapse in Multi-Agent RL**
Simon Yu, Nicholas Tomlin, Marwa Abdulhai, Ximing Lu, Derek Chong, Abe Hou, Dilara Soylu, Sergey Levine, Christopher D. Manning, Weiyan Shi
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026
41. **Can Aha Moments be Fake? Identifying True and Decorative Thinking Steps in Chain-of-Thought**
Jiachen Zhao, Yiyu Sun, Weiyan Shi[†], Dawn Song[†] ([†]equal advising)
Findings of the Association for Computational Linguistics: EMNLP, 2026
Coverage: [Quanta Magazine]
40. **Train Yourself as an LLM: Exploring Effects of AI Literacy on Persuasion via Role-Playing LLM Training**
Qihui Fan, Min Ge, Chenyan Jia, Weiyan Shi
Conference on Language Modeling (COLM), 2026
39. **Unsafer in Many Turns: Benchmarking and Defending Multi-Turn Safety Risks in Tool-Using Agents**
Xu Li, Simon Yu, Minzhou Pan, Yiyu Sun, Bo Li, Dawn Song, Xue Lin, Weiyan Shi
International Conference on Machine Learning (ICML), 2026
38. **Verbalized Sampling: How to Mitigate Mode Collapse and Unlock LLM Diversity**
Jiayi Zhang*, Simon Yu*, Derek Chong*, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, Weiyan Shi
International Conference on Machine Learning (ICML), 2026
Coverage: [Forbes] [Yahoo] [VentureBeat] [TLDR AI] [Neuron Daily]
37. **SPIRAL: Self-Play on Zero-Sum Games Incentivizes Reasoning via Multi-Agent Multi-Turn Reinforcement Learning**
Bo Liu*, Simon Yu*, Zichen Liu*, Leon Guertler*, Penghui Qi, Daniel Balcells, Mickel Liu, Cheston Tan, Weiyan Shi, Min Lin, Wee Sun Lee, Natasha Jaques
International Conference on Learning Representations (ICLR), 2026
36. **PolySkill: Learning Generalizable Skills Through Polymorphic Abstraction**
Simon Yu, Gang Li, Weiyan Shi[†], Peng Qi[†]
International Conference on Learning Representations (ICLR), 2026

35. **GEM: A Gym for Agentic LLMs**
 Zichen Liu, Anya Sims, Keyu Duan, Changyu Chen, Simon Yu, Xiangxin Zhou, Haotian Xu, Shaopan Xiong, Bo Liu, Chenmian Tan, Chuen Yang Beh, Weixun Wang, Hao Zhu, Weiyan Shi, Diyi Yang, Michael Shieh, Yee Whye Teh, Wee Sun Lee, Min Lin
International Conference on Learning Representations (ICLR), 2026
 Outstanding Paper, NeurIPS SEA Workshop, 2025
34. **LLMs Encode Harmfulness and Refusal Separately**
 Jiachen Zhao, Jing Huang, Zhengxuan Wu, David Bau, Weiyan Shi
Conference on Neural Information Processing Systems (NeurIPS), 2025
33. **Distilling an End-to-End Voice Assistant Without Instruction Training Data**
 William Held, Yanzhe Zhang, Minzhi Li, Weiyan Shi, Michael J. Ryan, Diyi Yang
Annual Meeting of the Association for Computational Linguistics (ACL), 2025
32. **NewsInterview: A Dataset and a Playground to Evaluate LLMs' Grounding Gap via Informational Interviews**
 Alexander Spangher, Michael Lu, Sriya Kalyan, Hyundong Justin Cho, Tenghao Huang, Weiyan Shi, Jonathan May
Annual Meeting of the Association for Computational Linguistics (ACL), 2025
31. **Proactive Conversational Agents with Inner Thoughts**
 Xingyu Bruce Liu, Shitao Fang, Weiyan Shi, Chien-Sheng Wu, Takeo Igarashi, Xiang Anthony Chen
ACM Conference on Human Factors in Computing Systems (CHI), 2025
30. **PrivacyLens: Evaluating Privacy Norm Awareness of Language Models in Action**
 Yijia Shao, Tianshi Li, Weiyan Shi, Yanchen Liu, Diyi Yang
Conference on Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, 2024
29. **Decoding Susceptibility: Modeling Misbelief to Misinformation Through a Computational Approach**
 Yanchen Liu, Mingyu Derek Ma, Wenna Qin, Azure Zhou, Jiaao Chen, Weiyan Shi, Wei Wang, Diyi Yang
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024
28. **Zero-Shot Persuasive Chatbots with LLM-Generated Strategies and Information Retrieval**
 Kazuaki Furumai, Roberto Legaspi, Julio Cesar Vizcarra Romero, Yudai Yamazaki, Yasutaka Nishimura, Sina Semnani, Kazushi Ikeda, Weiyan Shi, Monica Lam
Findings of the Association for Computational Linguistics: EMNLP, 2024
27. **CultureBank: An Online Community-Driven Knowledge Base Towards Culturally Aware Language Technologies**
Weiyan Shi, Ryan Li, Yutong Zhang, Caleb Ziems, Sunny Yu, Raya Horesh, Rogério Abreu de Paula, Diyi Yang
Findings of the Association for Computational Linguistics: EMNLP, 2024
26. **How Johnny Can Persuade LLMs to Jailbreak Them: Rethinking Persuasion to Challenge AI Safety by Humanizing LLMs**
 Yi Zeng*, Hongpeng Lin*, Jingwen Zhang, Diyi Yang, Ruoxi Jia[†], Weiyan Shi[†]
Annual Meeting of the Association for Computational Linguistics (ACL), 2024
 Best Social Impact Paper

25. **The Earth is Flat because...: Investigating LLMs' Belief towards Misinformation via Persuasive Conversation**
 Rongwu Xu, Brian S. Lin, Shujian Yang, Tianqi Zhang, Weiyan Shi[†], Tianwei Zhang, Zhixuan Fang, Wei Xu, Han Qiu[†]
Annual Meeting of the Association for Computational Linguistics (ACL), 2024
 Outstanding Paper
24. **Social Intelligence Data Infrastructure: Structuring the Present and Navigating the Future**
 Minzhi Li, Weiyan Shi, Caleb Ziems, Diyi Yang
Findings of the Association for Computational Linguistics: ACL, 2024
23. **A Safe Harbor for AI Evaluation and Red Teaming**
 Shayne Longpre, Sayash Kapoor, Kevin Klyman, Ashwin Ramaswami, Rishi Bommasani, Borhane Blili-Hamelin, Yangsibo Huang, Aviya Skowron, Zheng-Xin Yong, Suhas Kotha, Yi Zeng, Weiyan Shi, Xianjun Yang, Reid Southen, Alexander Robey, Patrick Chao, Diyi Yang, Ruoxi Jia, Daniel Kang, Sandy Pentland, Arvind Narayanan, Percy Liang, Peter Henderson
International Conference on Machine Learning (ICML), Position Track, 2024
22. **The Mirrored Influence Hypothesis: Efficient Data Influence Estimation by Harnessing Forward Passes**
 Myeongseob Ko, Feiyang Kang, Weiyan Shi, Ming Jin, Zhou Yu, Ruoxi Jia
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024
21. **When Life Gives You Lemons, Make Cherryade: Converting Feedback from Bad Responses into Good Labels**
Weiyan Shi, Emily Dinan, Kurt Shuster, Jason Weston, Jing Xu
Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2024
20. **Dialoging Resonance in Human-Chatbot Conversation: How Users Perceive and Reciprocate Recommendation Chatbot's Self-Disclosure Strategy**
 Kai-Hui Liang, Weiyan Shi, Yoo Jung Oh, Hao-Chuan Wang, Jingwen Zhang, Zhou Yu
Proceedings of the ACM on Human-Computer Interaction, CSCW, 2024
19. **AutoReply: Detecting Nonsense in Dialogue with Discriminative Replies**
Weiyan Shi, Emily Dinan, Adi Renduchintala, Daniel Fried, Athul Jacob, Zhou Yu, Mike Lewis
Findings of the Association for Computational Linguistics: EMNLP, 2023
18. **Controllable Mixed-Initiative Dialogue Generation through Prompting**
 Maximillian Chen, Xiao Yu, Weiyan Shi, Urvi Awasthi, Zhou Yu
Annual Meeting of the Association for Computational Linguistics (ACL), 2023
17. **Social Influence Dialogue Systems: A Survey of Datasets and Models For Social Influence Tasks**
 Kushal Chawla*, Weiyan Shi*, Jingwen Zhang, Gale Lucas, Zhou Yu, Jonathan Gratch
Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2023
16. **Just Fine-tune Twice: Selective Differential Privacy for Large Language Models**
Weiyan Shi, Ryan Shea, Si Chen, Chiyuan Zhang, Ruoxi Jia, Zhou Yu
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2022

15. **Seamlessly Integrating Factual Information and Social Content with Persuasive Dialogue**
Maximillian Chen, Weiyang Shi, Feifan Yan, Ryan Hou, Jingwen Zhang, Saurav Sahay, Zhou Yu
Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and International Joint Conference on Natural Language Processing (ACL-IJCNLP), 2022
14. **Towards Socially Intelligent Agents with Mental State Transition and Human Value**
Liang Qiu, Yizhou Zhao, Yuan Liang, Pan Lu, Weiyang Shi, Zhou Yu, Song-Chun Zhu
Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL), 2022
13. **Selective Differential Privacy for Language Modeling**
Weiyang Shi, Aiqi Cui, Evan Li, Ruoxi Jia, Zhou Yu
Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2022
12. **Refine and Imitate: Reducing Repetition and Inconsistency in Persuasion Dialogues via Reinforcement Learning and Human Demonstration**
Weiyang Shi, Yu Li, Saurav Sahay, Zhou Yu
Findings of the Association for Computational Linguistics: EMNLP, 2021
11. **LEGOEval: An Open-Source Toolkit for Dialogue System Evaluation via Crowdsourcing**
Yu Li, Josh Arnold, Feifan Yan, Weiyang Shi, Zhou Yu
Annual Meeting of the Association for Computational Linguistics (ACL), System Demonstrations, 2021
10. **PRAL: A Tailored Pre-Training Model for Task-Oriented Dialog Generation**
Jing Gu, Qingyang Wu, Chongruo Wu, Weiyang Shi, Zhou Yu
Annual Meeting of the Association for Computational Linguistics (ACL), 2021
9. **INSPIRED: Toward Sociable Recommendation Dialog Systems**
Shirley Anugrah Hayati, Dongyeop Kang, Qingxiaoyang Zhu, Weiyang Shi, Zhou Yu
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020
8. **Structured Attention for Unsupervised Dialogue Structure Induction**
Liang Qiu, Yizhou Zhao, Weiyang Shi, Yuan Liang, Feng Shi, Tao Yuan, Zhou Yu, Song-Chun Zhu
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020
7. **Understanding User Resistance Strategies in Persuasive Conversations**
Youzhi Tian, Weiyang Shi, Chen Li, Zhou Yu
Findings of the Association for Computational Linguistics: EMNLP, 2020
6. **Effects of Persuasive Dialogues: Testing Bot Identities and Inquiry Strategies**
Weiyang Shi, Xuwei Wang, Yoo Jung Oh, Jingwen Zhang, Saurav Sahay, Zhou Yu
ACM Conference on Human Factors in Computing Systems (CHI), 2020
5. **End-to-End Trainable Non-Collaborative Dialogue System**
Yu Li, Kun Qian, Weiyang Shi, Zhou Yu
AAAI Conference on Artificial Intelligence (AAAI), 2020
4. **How to Build User Simulators to Train RL-based Dialog Systems**
Weiyang Shi*, Kun Qian*, Xuwei Wang, Zhou Yu
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2019

3. **Persuasion for Good: Towards a Personalized Persuasive Dialogue System for Social Good**
Xuewei Wang*, Weiyang Shi*, Richard Kim, Yoo Jung Oh, Sijia Yang, Jingwen Zhang, Zhou Yu
Annual Meeting of the Association for Computational Linguistics (ACL), 2019
Best Paper Nomination
2. **Unsupervised Dialog Structure Learning**
Weiyang Shi, Tiancheng Zhao, Zhou Yu
Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2019
1. **Sentiment Adaptive End-to-End Dialog Systems**
Weiyang Shi, Zhou Yu
Annual Meeting of the Association for Computational Linguistics (ACL), 2018

PREPRINTS

5. **SPADE: Self-Play in Adaptive Synthetic Executable Environments**
Bo Liu*, Simon Yu*, Yiding Jiang, Ao Qu, Andrew Zhao, Zichen Liu, Junsu Kim, Zijian Zhou, Seungone Kim, Tongzheng Ren, Mickel Liu, Hanfei Yu, Zhaorun Chen, Weiyang Shi, Paul Pu Liang, Luke Zettlemoyer, Yejin Choi, Natasha Jaques
arXiv preprint, 2026
4. **Agents' Last Exam**
Yiyu Sun et al. (multi-institution collaboration led by Berkeley RDI)
arXiv preprint, 2026
3. **Shepherd: Enabling Programmable Meta-Agents via Reversible Agentic Execution Traces**
Simon Yu*, Derek Chong*, Ananjan Nandi*, Dilara Soylu, Jiuding Sun, Christopher D. Manning, Weiyang Shi
arXiv preprint, 2026
2. **Coding with "Enemy": Can Human Developers Detect AI Agent Sabotage?**
Jingheng Ye*, Huiqi Zou*, Simon Yu, Weiyang Shi
arXiv preprint, 2026
Best Paper, ICML DL4C Workshop, 2026
1. **The Piggyback Hypothesis of Generalization: Explaining and Mitigating Emergent Misalignment**
Jiachen Zhao, Zhengxuan Wu, Aryaman Arora, Yiyu Sun, David Bau, Weiyang Shi
arXiv preprint, 2026

FUNDING

Total funding: \$1.9M as single-PI (\$210K anticipated), \$2.3M overall

Epistemic Sampling for Meta-Calibration (anticipated) 2026/10 – 2027/10
Coefficient Giving, Single-PI, \$210K

Self-Exploring Safety Layer for Physical Environments 2026/06 – 2027/06
Samsung, Single-PI, \$125K

Decoding Agentic Persuasion: Understanding and Monitoring Persuasion by AI Agents to Mitigate Harm 2025/11 – 2028/11
Schmidt Sciences AI2050 Early Career Fellowship, Single-PI, \$500K

	Mitigating Emergent Misalignment Coefficient Giving, Single-PI, \$330K	2025/10 – 2027/03
	Agentic Sabotage Capability Evaluation Coefficient Giving, Single-PI, \$707K	2025/08 – 2027/08
	Leveraging Generative AI for Updating Coastal Risk Assessments Massachusetts AI Hub, Co-PI, \$200K out of \$500K	2025/10 – 2027/10
	PATCH of Asclepius: Technology for Cybersecure Healthcare ARPA-H, Co-I, \$200K out of \$14M	2025/09 – 2026/09
MENTORSHIP	Ph.D. Simon Yu (<i>Best Paper, ICML DL4C 2026; Outstanding Paper, NeurIPS SEA 2025</i>) Jiachen Zhao (<i>MATS scholar</i>) Jingheng Ye (<i>Best Paper, ICML DL4C 2026</i>) Huiqi Zou (<i>Best Paper, ICML DL4C 2026</i>) Mingqi Gao Xu Li Barrett Lattimer	2024 – Present 2024 – Present 2025 – Present 2025 – Present 2025 – Present 2026 – Present 2026 – Present
	Master's Qihui Fan (<i>first paper at COLM; → Ph.D., Stony Brook University</i>) Karmen Cai (<i>→ Amazon</i>)	2025 – 2026 2025 – 2026
	Undergraduate Xu Li (<i>first paper at ICML; → Ph.D., Northeastern University</i>) Miranda Anna Christ Julius Broomfield	2025 – 2026 2026 – Present 2026 – Present
	Other Mentees Rongwu Xu (<i>Outstanding Paper, ACL; → Ph.D., University of Washington</i>)	2023 – 2024
INVITED TALKS	<i>Git for Agents: Rewind, Fork, and Audit Your Agents with Meta-Agents.</i> Human-Centered Machine Learning Workshop, Apple, 2026/08. <i>Beyond the Surface: How Post-Training Artifacts Shape LLM Diversity and Safety.</i> University of Pennsylvania NLP Seminar, 2026/03; Future of Life Institute, 2026/02; Stanford University NLP Seminar, 2025/10. <i>Persuasion for Good: How to Build and Break Chatbots.</i> Alignment Workshop, 2025/04; AAAI New Faculty Highlights, 2025/02; Google Cloud AI, 2024/11; Capital One AI, 2024/11; Tufts University, 2024/10; University of Massachusetts Lowell Colloquium, 2024/10; Meta Llama Team, 2024/10; Amazon Research, 2024/10; Adobe Research, 2024/09.	

Interactive AI Systems Specialized in Social Influence. Carnegie Mellon University, 2024/02; University of Southern California, 2023/10; KAIST, 2023/09; Stanford University NLP Seminar, 2023/06; New York University, 2023/05; North Carolina State University, 2023/05; Cornell Tech, 2023/04.

TEACHING	ECE 5668, Large Language Models	Fall 2026
	<i>(designed this course)</i>	
	ECE 7398, Large-Language-Model-based Agents	Spring 2026
	<i>(Effectiveness Mean=4.6, Median=5.0)</i>	
	ECE 7398, Large-Language-Model-based Agents	Spring 2025
	<i>(Effectiveness Mean=4.7, Median=5.0)</i>	
	CS 7180, Large-Language-Model-based Agents	Fall 2024
	<i>(designed this new course; Effectiveness Mean=3.6, Median=4.0)</i>	
SERVICE	Workshop Organizer: 1st Workshop on Responsible Use of Meta-Agents (NeurIPS 2026); 1st Workshop on Multi-Turn Interactions in LLMs (NeurIPS 2025); 1st Workshop on Advancing LLM-Based Multi-Agent Collaboration (AAAI 2025); 2nd Workshop on Social Influence in Conversations (EMNLP 2024); 1st Workshop on Social Influence in Conversations (ACL 2023); 4th Workshop on NLP for Conversational AI (ACL 2022). Area Chair: WiML 2022; ACL 2023–2026; EMNLP 2023–2026; NAACL 2023–2025; ICML 2026; NeurIPS 2026. Conference Reviewer: ACL 2019–2022; ACL Rolling Review 2021–2022; *SEM 2020–2021; NAACL 2021; EACL 2021; EMNLP 2021; SIGDIAL 2022; NLPCC 2021; COLM 2025; ICLR 2021–2022; ICML 2022; NeurIPS 2021; AAAI 2021–2022; CHI 2022. Journal Reviewer: ACM Transactions on Human-Robot Interaction; Neurocomputing; ACM Transactions on Information Systems; Nature Human Behaviour; PNAS Nexus.	